

Education

- M.Sc in Environmental Change and Global Sustainability | 2025-**
University of Milan, Milano, Italy.
- M.Sc in Mining Exploration Engineering | 2020 - 2022**
Amirkabir University of Technology, Tehran, Iran.
GPA: 3.63/4 (16.89/20)
Thesis Topic: Mineral Potential Detection in Remote Sensing and GIS Environment.
- B.Sc. in Mining Engineering | 2015 - 2020**
Shahid Bahonar University of Kerman (SBU), Kerman, Iran.
Thesis Topic: Time-lapse ERT Monitoring of Tailings Dams. (Thesis Grade: 4/4, (19.5/20))
Thesis Advisor: Dr. Azadeh Hojat

RESEARCH INTERESTS

- Sustainability-focused Earth Sciences
- Environmental Geochemistry
- Sustainable Mining Practices
- Sustainability Analysis
- Critical Raw Materials Management

HONORS & AWARDS

- Ranked 11th among 589 participants in the National University Entrance Exam for M.Sc. degree in Mining and Environment Engineering in Iran. 2020
- Ranked 19th among 592 participants in the National University Entrance Exam for M.Sc. degree in Mining Exploration Engineering in Iran. 2020
- Ranked within the top 10% of Mining Engineering undergraduate students, Shahid Bahonar University of Kerman, Iran. 2020

TEST SCORES

- TOEFL Overall: 93
- GRE: 304 (Verbal 147, Quant 157, Analytical Writing 3)

PUBLICATIONS

- Mahmoudmolaei Kermani, A.,** Pearl, L.; **Climate Change effects on Maiz in Etupia;** (In prep.)

CONFERENCES

- Mahmoudmolaei Kermani, A.,** Hassani, H.; **"Detection of Cu element alterations using Landsat 8 images in Nagisan region, Kerman";** Fifth International Conference on Technology Development in Materials Engineering, Mining and Geology, Iran, Feb 19, 2022. <https://civilica.com/doc/1425988/>
- Mahmoudmolaei Kermani, A.,** Hassani, H.; **"Mineral potential detection using AHP and SWARA methods in Nagisan area, east of Kerman",** Sixth International Conference on Technology Development in Materials Engineering, Mining and Geology. Iran, 2022. <https://civilica.com/doc/1535538/>

BOOKS

- Mosavi Gheih Ghashlaghy, S. M., Kordpour, I., & **Mahmoudmolaei Kermani, A.** (2024). **Geotechnical monitoring with accurate and advanced tools in crisis management.** [Ebadi Publication].

COMPUTER SKILLS

- **Programming Languages:** Python (Geopandas, matplotlib), R
- **Applied Software:** Envi, Datamine, ArcGIS, Dips, Datamine, Surfer, SPSS, Stereonet

ACADEMIC PROJECTS

Impacts of Mining on Bird Populations (Val Sora Quarry) | *University of Milan,*

- Designed an ecological monitoring protocol using line transects to assess the environmental impacts of mining on local bird communities in the Italian Alps.
- Utilized GIS tools to map spatial data, establish buffer zones around the quarry, and design systematic survey transects.

LABORATORY SKILLS

- **Soil mechanics laboratory:** Sorting, preparing, and performing needed analyses, like CBR and Sieve analysis
- **X-ray laboratory for cement:** Taking samples, grinding and powdering, and analysing them with XRF
- **Descriptive mineralogy laboratory:** Working with polarizing microscope

WORK EXPERIENCE

- **Teaching English,** May 2024
- **Intern,** June 2020 – August 2020
Technical and Soil Mechanics Lab. co, Kerman, Iran.
Description: Performed various soil and rock mechanics experiments.
- **Intern,** June 2019 – August 2019
Momtazan Cement Factory, Kerman, Iran.
Description: Worked in chemistry, rock mechanics, and X-ray labs; acquainted with vehicles and methods of mining.

SELECTED COURSES

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| • Advanced Structural Geology: | 4/4 | • Advanced Engineering Mathematics: | 4/4 |
| • Design of Geochemical Exploration Projects: | 4/4 | • CLIMATE CHANGE: IMPACT AND ADAPTATION: | 4/4 |
| • Principles of Geochemical Exploration: | 4/4 | • Remote Sensing and GIS: | 4/4 |

REFERENCES

- Dr. Azadeh Hojat, Associate Professor, Department of Mining Engineering, Shahid Bahonar University of Kerman, Kerman, Iran.
Email: ahojat@uk.ac.ir
- Dr. Amirhossein Najafabadipour, Assistant Professor, Department of Mining Engineering, University of Jiroft, Iran.
Email: najafabadipour@ujiroft.ac.ir